Decoding Pre-Response EEG Signatures of Choice, Accuracy, and Confidence in Perceptual Decision-Making

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Abstract: Scalp EEG may carry information about an upcoming decision, but it is unclear how much of it is independent of response time and whether it can be used beyond a retrospective analysis. Using a public simultaneous EEG-fMRI random-dot motion dataset (24 participants, 7,377 trials), we decoded choice, accuracy, and confidence from a 600 ms window ending about 150 ms before the response, with out-of-participant validation and permutation tests. Mean AUC was 0.527, 0.535, and 0.540, all above chance. Accuracy and confidence effects vanished after flexible response-time control, and adding EEG to response time did not improve prediction. The choice effect survived response-time control but not matching on response time and stimulus direction. Results were stable across normalization, causal filtering, and classifiers. High-confidence errors were not separable from low-confidence errors, and decoded confidence did not predict next-trial behaviour. The findings describe weak, retrospective information.

Keywords: Perceptual decision-making, Electroencephalography, Decision confidence, Single-trial decoding, Response time control, Cross-validation

1. Introduction

Three claims about EEG and decisions are often conflated: that EEG activity is related to a person's choice or confidence, that it allows these to be predicted on a single trial, and that it causes them. Observational EEG supports at most the first two, and the third is outside the scope of this study. The dataset analysed here [1–2] already supports the first claim: a stimulus-independent EEG component that discriminates confidence appears before the explicit report [2]. We therefore pursue prediction, in four research questions. **RQ1:** can pre-response EEG predict the upcoming left or right judgment? **RQ2:** can it predict whether the judgment will be correct? **RQ3:** can it distinguish the joint states of accuracy and confidence, in particular high-confidence errors from low-confidence errors? **RQ4:** does the neural confidence state on trial N predict behaviour on trial N+1?

Evidence linking EEG to error detection and confidence comes largely from components that follow the response, such as the error positivity [3–4], which cannot predict a decision before it is made. Activity before the response has received less attention. A second distinction concerns the kind of prediction. Here the window is defined from each trial's own response time, so its endpoint cannot be located before the response; this is *retrospective decoding* of pre-response activity and not *online prediction*. We keep the two apart, and Section 5.6 adds a stimulus-locked evaluation that uses only information available at a fixed time after stimulus onset.

Our contributions are: (i) a response-buffered window with documented temporal support of the pre-processing; (ii) leave-one-subject-out decoding of choice, accuracy, and confidence with permutation inference; (iii) response-time controls that include flexible adjustment, incremental tests against response-time-only models, and trial matching; (iv) comparisons with alternative decoders and component baselines using equivalence tests; (v) small-sample-corrected calibration and sequential analyses with a sensitivity analysis for every null claim; and (vi) an account of the analysis plan, which was internal, amended, and not preregistered (Section 4.5). The results are modest, and we report them as such.

2. Related Work

A centro-parietal positivity (CPP) has been characterized as a domain-general decision variable that builds up to a threshold before a perceptual report [5]. In a random-dot motion task its build-up rate scaled with evidence strength and predicted reaction time within coherence levels [6]. Because the CPP indexes the unsigned strength of accumulated evidence [7], decoding the side of a choice must also draw on signals that differ between alternatives, including motor preparation [8]. Single-trial linear classifiers over EEG components were established by Philiastides et al. [9] and extended by Ratcliff et al. [10], and the same framework separates correct from incorrect trials [9]. Free-choice paradigms decoded choices from fMRI seconds before awareness [11–12], but they differ from evidence-driven discrimination.

Post-response components relate to confidence: the error positivity varies with confidence independent of accuracy [3], may reflect post-decisional evidence accumulation [4], and is partly distinct from other markers of confidence [13]. Confidence on one trial shifts decision bounds on the next [14], and choice history biases follow confidence-modulated accumulation [15–16]. We use "overconfidence" operationally, as a high-confidence error, which is coarser than measures of metacognitive sensitivity [17].

Three studies are closest to ours. Dou et al. [18] found that the slope of a pre-decision centro-parietal positivity predicted confidence with accuracy, response time, and evidence strength controlled, using participant-level regression on one component. Balsdon and Philiastides [19] applied a decoder trained on post-response activity to pre-response EEG to obtain an online confidence signal within participants. Fernandez-Vargas et al. [20] predicted continuous confidence from EEG and response time across eight experiments, and response time helped within participants but hurt transfer. These studies differ from ours in features, normalization, task, and target, so AUC values cannot be ranked across studies. Instead, we evaluate a component baseline (CPP-like and lateralized-readiness-like features) on the same trials and splits (Section 5.5). Small decoding effects in small samples can also exceed analytic chance levels [21], which is why we use permutation inference.

3. Dataset and Task

We used the open dataset of Gherman and Philiastides [1–2] (OpenNeuro ds002739, CC0 license): 24 participants, two runs each, recorded with simultaneous EEG-fMRI during a speeded left or right random-dot motion discrimination at an individually calibrated coherence (targeting about 75% accuracy). After the response, participants rated confidence on a 9-point scale whose on-screen orientation was randomized across trials to decorrelate motor preparation from the report. We used the pre-processed EEG supplied with the dataset. Its README lists the operations (scanner-gradient artifacts removed by template subtraction, a 12 ms median filter, 0.5 to 40 Hz band-pass filtering, resampling to 1000 Hz, principal-component removal of ocular and ballistocardiogram components, and baseline correction over the 100 ms before stimulus onset) but does not state the type, order, or phase of the filters; their temporal support is discussed in Section 4.1. Low confidence was defined as ratings 1 to 3 and high confidence as ratings 7 to 9, following the dataset authors' split into the lowest and highest three ratings; ratings 4 to 6 were labeled medium and are excluded from the confidence analyses only. The 53 channels present in every participant were used, and consecutive trials were identified through the original trial index so that trial pairs in the sequential analysis are truly adjacent. Motion direction and choice were coded so that the recorded accuracy agrees with the mapping of direction to response on all 7,377 valid trials.

4. Methods

4.1 Window definition and temporal support

Pre-processed EEG was decimated from 1000 Hz to 250 Hz with a zero-phase Chebyshev type I filter (order 8, cut-off 100 Hz). For every trial a window of length L ends a buffer B before that trial's own response time t_{resp}. The mean amplitude of channel c is

$$x_c = (1 / n_w) \sum_{t=t0}^{t1-1} e_c(t) \qquad (1)$$

where $t_0 = t_{\text{resp}} - B - L$, $t_1 = t_{\text{resp}} - B$, and n_w is the number of samples. A trial is retained if RT $\geq L + B$. Because epochs were cut at 250 Hz, the realized window spans about −755 to −152 ms relative to the response (Fig. 1).

Selection rule. The grid was L from 200 to 800 ms and B from 0 to 350 ms (56 windows; Supplementary Table S2). Among windows with $B \geq 100$ ms and at least 80% retention we chose the largest L, and among ties the largest B. For $L = 600$ ms the buffers 100 and 150 ms both qualify (86.2% and 82.6% of trials), so the tie-break selects $B = 150$ ms. The rule used retention only and never decoding

performance, and the tie-break was in the internal plan but missing from the first manuscript.

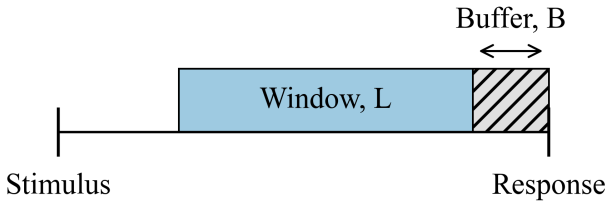


Fig. 1. Schematic of the response-buffered window: a window of length L ends a buffer B before the response, so its endpoint depends on the trial's own response time (retrospective decoding).

Temporal support. A 150 ms buffer alone does not show isolation from response-related activity. The dataset's template subtraction, median filter, band-pass filter, and component removal are only partly documented, and the raw data are not released, so we cannot show that none of them uses later samples; the protocol is therefore not called leakage-controlled. We characterized the operations we apply (analytic kernels, Supplementary Table S9) and re-extracted the features from the 1 kHz data with a plain mean, causal decimation, and decimation of data truncated at the window end. As a proxy for the undocumented band-pass we also applied a 0.5–40 Hz Butterworth filter in zero-phase, causal, and truncated form (Section 5.4).

4.2 Decoder, target coding, and normalization

The window-averaged channel vector $\mathbf{x}$ is projected onto a weight vector $\mathbf{w}$ to give one score per trial,

$$s = \mathbf{w}^{\mathrm{T}}\mathbf{x} + b \tag{2}$$

with weights from class-weighted L2 logistic regression (inverse class-frequency weights, unpenalized intercept, $C = 0.1$ fixed in advance; $C = 0.01$ and 1 as checks), solved exactly by Newton iterations to a gradient tolerance of 10^{-9} and validated against scikit-learn (Supplementary Table S5). The positive class is a right-hand choice, a correct response, or a high rating (7 to 9; ratings 1 to 3 form the negative class and 4 to 6 are excluded from the confidence target only); larger scores indicate the positive class. All fitting used training participants only.

Normalization. Each feature was standardized within participant over that participant's retained trials (both runs, no labels). This is transductive adaptation to an unlabelled batch of the held-out participant and not label leakage, but it requires a batch of the new participant's trials and, for the RT control, completed response times. Trials excluded by the window rule enter neither normalization nor training nor testing, and ratings 4 to 6 contribute to normalization only. As sensitivity analyses we standardized with all valid trials, with training participants only, with a separate calibration run, and with causal running statistics, and we estimated the RT regression on training participants only (Section 5.5). The first manuscript mixed the all-trial and retained-trial batches, which explains its third-decimal discrepancies.

Comparison decoders. We compared the primary decoder with shrinkage LDA, a linear SVM, an RBF SVM, a random forest, and a neural network, and with sub-bin means and log variance as features, all on the same trials and splits with hyperparameters fixed in advance (Supplementary Table S4). Equivalence was tested against a margin of ±0.02 AUC.

4.3 Validation, inference, and response-time control

Cross-validation is leave-one-subject-out (LOSO). We report the area under the ROC curve,

$$\mathrm{AUC} = \mathrm{P}(s^{+} > s^{-}) + 0.5\,\mathrm{P}(s^{+} = s^{-}) \tag{3}$$

computed per held-out participant (participants with at least 10 trials and 3 per class) and for the pooled scores. Chance was estimated by permutation: labels were shuffled within participant and the whole LOSO procedure repeated 1000 times, with $p = (1 + k) / (1 + 1000)$. The primary statistic is the mean participant AUC, the three primary tests were Holm-corrected, and intervals are percentile bootstrap intervals over participants.

Response-time control Section 5.3. Removing an RT association from the EEG is not a causal demonstration of speed contamination, because RT shares the decision process with the neural features and adjustment can remove genuine information. We therefore report five analyses that ask different questions: RT only (linear and a 5-degree-of-freedom natural spline); residualization of the EEG features,

$$\tilde{\mathbf{x}} = \mathbf{x} - \boldsymbol{\beta}\, g(\mathrm{RT}) \tag{4}$$

with $g(\mathrm{RT})$ the linear or spline basis and $\boldsymbol{\beta}$ the least-squares coefficients; incremental prediction (EEG plus flexible RT against RT only, tested by permuting labels within participant and RT decile); stratification by participant-specific RT terciles; and matching trials within participant, RT decile, and stimulus direction.

4.4 Additional analyses

Components and time course. CPP-like (mean amplitude and slope over CPz, CP1, CP2) and LRP-like (C3/C4, C5/C6, CP3/CP4, CP5/CP6 differences) baselines were evaluated on the same trials and splits. The decoding was repeated in 100 ms windows between −750 and −50 ms, and in a stimulus-locked setting with a 300 ms window ending at 400 to 1100 ms after onset and a processing latency of 0 or 150 ms.

Calibration (RQ3). Accuracy was crossed with low and high confidence, giving four cells. For each contrast we report class counts per held-out participant, the number of evaluable participants (both classes present), the mean and pooled AUC, participant-bootstrap intervals, and permutation p-values (labels shuffled within participant inside the contrast; whole procedure re-fitted). A four-way classifier and a two-stage analysis (accuracy-decoder score of incorrect/high against incorrect/low trials) were added. The feasibility rule (at least 100 incorrect/high trials and no participant above 25%) shows that the analysis is possible and not that it is powered.

Sequential effects (RQ4). For consecutive retained trials N and N+1 in a run, the neural confidence state S is the out-of-participant high-versus-low decoder score on trial N (z-scored within participant) and C is reported confidence, used as a benchmark. Choice repetition was modeled by a logistic GEE with an exchangeable working correlation,

$$\text{logit } P(\text{repeat}) = \beta_0 + \beta_1 S + \beta_2 D + \beta_3 A + \beta_4 T \quad (5)$$

where D indicates a repeated motion direction, A is accuracy on trial N, and T is ln RT on trial N. Log RT and accuracy on trial N+1 were modelled with identity and logit links,

$$\ln RT_{N+1} = \beta_0 + \beta_1 S + \beta_2 T + \beta_3 A; \text{ logit } P(\text{accurate}_{N+1}) = \beta_0 + \beta_1 S + \beta_2 A + \beta_3 T \quad (6)$$

S is a generated regressor, so with 24 clusters we report robust, bias-reduced (Mancl and DeRouen [22], t on 23 degrees of freedom) inference and a participant bootstrap (500 resamples) that re-fits the decoder in every resample.

Sensitivity. A non-significant result is not evidence of absence. For every null claim we report an interval and the minimum detectable effect at 80% power (Hanley and McNeil [23] for pooled contrasts; permutation null plus 0.84 standard errors for mean AUCs). We take AUC 0.55 as the smallest effect of interest and ±0.02 as the equivalence margin [24–25]; the permutation 95th percentile is a rejection threshold and not a power estimate.

4.5 Plan provenance and reproducibility

The analyses followed an internal plan document that has no external time stamp and was edited in place. File-system times (Supplementary Table S1) show that the plan was created before the full-sample run, that the stimulus-locked, time-resolved, and tercile analyses came later, and that the comparison decoders were added as a second amendment. The plan was not preregistered. The flexible RT control, matching, component baselines, and all analyses added for this revision are post hoc. A versioned package with the plan, environment, channel mask, trial and pair indices, seeds, predictions, permutation statistics, and table-generation scripts accompanies the paper.

5. Results

5.1 Behaviour and retention

The analysis includes 24 participants, 48 runs, and 7,377 valid trials. Accuracy averaged 76.0% (SD 5.2), close to the calibrated target, and mean RT was 992 ms. Confidence was higher on correct than on incorrect trials (5.73 versus 4.33), and RT was negatively related to confidence (mean within-participant Spearman ρ = −0.26) and to accuracy (ρ = −0.18). The retained trials are slower, less accurate (74.6% versus 82.4%), and less confident (5.21 versus 6.28) than the 1,287 excluded trials, so the decoding results apply to these slower trials. Retention was very uneven across participants: the median participant retained 97.3% of valid trials, but 4 participants retained less than half and participant 22 only 24.2%, so participants contributed between 77 and 317 retained trials (median 298; Supplementary Table S3). The results therefore describe mostly the slower responders' slower trials and should not be extended to fast decisions. The incorrect/high cell contains 237 trials in the analysis window, contributed by 23 of 24 participants with no participant contributing more than 18.1%.

Table 1. Dataset and behavior

Measure	Value
Participants / runs	24 / 48
Valid trials	7,377
Valid trials per participant	290–318
Accuracy, mean (SD)	76.0% (5.2)
RT, mean (SD)	992 (239) ms
Trials in window	6,090 (82.6%)
Retained per participant, median (range)	298 (77–317)
Retention per participant, median (min)	97% (24%)
Channels used	53

5.2 RQ1 and RQ2: choice, accuracy, and confidence

Out-of-participant decoding of choice was above chance (mean AUC 0.527, SD 0.041, 95% interval [0.511, 0.543]; 18 of 24 participants above 0.5; permutation $p = 0.005$), as was accuracy (0.535, SD 0.047, [0.517, 0.554]; 17 of 24; $p = 0.003$) and confidence, high versus low (0.540, SD 0.066, [0.515, 0.566]; 16 of 24; $p = 0.006$) (Table 2). The permutation null had a mean of 0.500 and a 95th percentile of 0.518 for choice (0.519 for accuracy and 0.528 for confidence). With Holm correction across the three primary tests the adjusted p-values were 0.010, 0.009, and 0.010. Results were insensitive to the regularization strength and to shrinkage LDA (choice 0.525, accuracy 0.536, confidence 0.541). Within-participant cross-validation yielded 0.550 for choice, 0.541 for accuracy, and 0.583 for confidence, higher than the out-of-participant values, which suggests that part of the information is participant-specific.

Table 2. Primary decoding performance (AUC, retrospective window)

	Choice	Accuracy	Confidence
LOSO mean (SD)	0.527 (0.041)	0.535 (0.047)	0.540 (0.066)
95% interval	[0.511, 0.543]	[0.517, 0.554]	[0.515, 0.566]
Participants above 0.5	18/24	17/24	16/24
Permutation p	0.005	0.003	0.006
Holm-adjusted p	0.010	0.009	0.010
Null 95th percentile	0.518	0.519	0.528
LOSO pooled AUC	0.522	0.540	0.548
Shrinkage LDA	0.525	0.536	0.541
$C = 0.01$ / 1	0.524 / 0.527	0.537 / 0.535	0.544 / 0.541
Within-participant 10-fold	0.550	0.541	0.583

5.3 Response-time control

Response time alone predicted accuracy (AUC 0.614 with the flexible RT term) and confidence (0.688) better than the EEG did, but not choice (0.505); the linear RT term gave 0.615, 0.688, and 0.508. After removing the linear dependence of the EEG features on RT, choice decoding was unchanged (AUC 0.526, $p = 0.009$), whereas accuracy decoding fell to chance (0.503, $p = 0.419$) and so did confidence decoding (0.512, $p = 0.231$); the flexible spline gave 0.527 ($p = 0.005$), 0.502 ($p = 0.462$), and 0.504 ($p = 0.405$). A decoder with EEG and flexible RT predicted accuracy no better than RT alone (AUC 0.604 versus 0.614; difference −0.010, [−0.021, −0.001]; RT-stratified permutation $p = 0.169$) or confidence (0.667 versus 0.688; −0.021, $p = 0.776$), and the point estimates are negative because the additional features add estimation noise in out-of-participant transfer. For choice the EEG added +0.019 over RT alone (0.524 versus 0.505; participant-level interval [−0.003, 0.041], improvement in 14 of 24 participants; RT-stratified permutation $p = 0.010$). Stratifying by RT without a linear model gave a mean AUC within participant-specific RT terciles of 0.528 for choice (permutation $p = 0.005$; one-sided t-test on participant AUCs $p = 0.005$), 0.515 for accuracy (permutation $p = 0.106$; t-test $p = 0.056$), and 0.496 for confidence (permutation $p = 0.592$; t-test $p = 0.590$). The first version of the manuscript reported a significant accuracy effect within terciles (AUC 0.517, $p = 0.030$) from a t-test on a version of the scores that was normalized differently; with the single normalization used here neither test is significant, so the accuracy result after stratification is borderline rather than clearly present, and coarse terciles keep within-bin RT variation, which is why the flexible and incremental analyses above are preferred. For choice the AUC was highest in the fastest tercile (0.549, versus 0.522 and 0.513), compatible with a larger motor or evidence contribution close to the response. Matching trials on RT changed the choice result. When choice classes were equalized within participant, RT decile, and stimulus direction, the choice AUC was 0.506 (SD 0.008 over 10 draws; 2,772 trials per draw; permutation $p = 0.292$), and matching on RT decile alone for accuracy and confidence gave 0.496 ($p = 0.618$) and 0.505 ($p = 0.664$). The matched choice analysis uses fewer trials, so its failure to reach significance does not show that the choice information is absent (Section 5.10 gives its sensitivity), but the evidence that the choice effect is independent of RT and stimulus direction jointly is weak.

Table 3. Response-time control: uncorrected and corrected inference (mean participant AUC, permutation *p*)

Analysis	Choice	Accuracy	Confidence
Uncorrected	0.527 (0.005)	0.535 (0.003)	0.540 (0.006)
RT only, linear	0.508	0.615	0.688
RT only, flexible	0.505	0.614	0.688
Residualized, linear	0.526 (0.009)	0.503 (0.419)	0.512 (0.231)
Residualized, flexible	0.527 (0.005)	0.502 (0.462)	0.504 (0.405)
EEG + RT, flexible	0.524	0.604	0.667
Increment over RT only	+0.019 (0.010)	−0.010 (0.169)	−0.021 (0.776)
RT-tercile stratified	0.528 (0.005)	0.515 (0.106)	0.496 (0.592)
RT-matched	0.506 (0.292)	0.496 (0.618)	0.505 (0.664)

5.4 Temporal support of the pre-processing

A zero-phase 0.5–40 Hz Butterworth band-pass, in contrast, places 32% of its weight after the window end, 23% beyond 150 ms. If the band-pass supplied with the dataset is a zero-phase filter of this kind, samples after the buffer can in principle enter the window. This study cannot exclude it, because the filter is not documented and the raw recordings were not available in unfiltered form. The empirical check could be completed for the 9 participants whose two runs were re-extracted from the 1 kHz recordings (2,265 retained trials). Replacing the decimated mean by a plain mean of the 1 kHz samples, by causal decimation, or by decimation of data truncated at the window end changed the AUCs by at most 0.007 on average and by at most 0.024 for any participant (Table 4). The decimation step therefore does not carry the result. Causal filtering shifts the phase of every component and truncation creates an edge transient, so a lower AUC under these variants can reflect the change of the signal itself rather than leakage. The causal and the truncated band-pass changed the accuracy AUC by −0.067 and −0.070, and the truncated band-pass changed the choice AUC by −0.040, while the correlations between these features and the primary features were only 0.16 and 0.26. Such large feature changes show that the proxy variants alter the signal drastically, and they do not isolate leakage. We therefore state the result as follows: the decimation step is not responsible for the effects, the effect of the dataset's own band-pass remains untested, and the protocol is not described as leakage-controlled.

Table 4. Change in mean AUC relative to the primary features for variants of the temporal processing, with 95% intervals

Variant	Choice	Accuracy	Confid.
Raw 1 kHz mean (no decimation filter)	+0.001 [−0.000, 0.001]	−0.000 [−0.002, 0.001]	+0.000 [−0.001, 0.001]
Causal decimation filter	−0.002 [−0.004, 0.001]	+0.001 [−0.003, 0.003]	−0.007 [−0.014, 0.001]
Decimation of data truncated at window end	−0.000 [−0.000, 0.000]	+0.000 [−0.000, 0.000]	−0.000 [−0.000, 0.000]
Added 0.5–40 Hz band-pass, zero phase	−0.005 [−0.016, 0.003]	−0.010 [−0.015, −0.004]	+0.001 [−0.005, 0.008]
Added band-pass, causal	−0.004 [−0.040, 0.027]	−0.067 [−0.098, −0.027]	−0.022 [−0.085, 0.038]
Added band-pass, zero phase, truncated at window end	−0.040 [−0.065, −0.018]	−0.070 [−0.096, −0.042]	+0.001 [−0.033, 0.034]

5.5 Normalization, decoders, components, and windows

Standardizing with the first version's all-trial batch changed the mean AUC by at most 0.0003 (this is the source of the third-decimal differences between the tables of the first version), standardizing with statistics of the pooled training participants only (no target-participant data) by −0.0031, +0.0023, and +0.0031 for choice, accuracy, and confidence, and using the participant's other run as an unlabeled calibration batch by −0.0021, −0.0005, and −0.0043; all intervals of these differences include zero. Causal running standardization (only earlier trials of the same run, which removes the first 20 trials of each run) gave AUCs of 0.513, 0.544, and 0.540; the choice value was lower than the canonical one (−0.0134, [−0.028, −0.000]), which may reflect the loss of trials or the noisier running statistics. These analyses show robustness of the AUCs to the normalization choice; they do not turn the protocol into a prospective test in a new participant, because the retained-trial rule and the RT control still require completed responses. Shrinkage LDA and the linear SVM were equivalent to the primary decoder within ±0.02 AUC for every target (differences −0.0012 to +0.0014), whereas nonlinear classifiers and spatiotemporal features gave larger differences in both directions with intervals that neither excluded the margin nor showed an advantage after correction (Supplementary Table S4). Log-variance features produced AUCs below 0.5 for accuracy and confidence (0.476 and 0.471), which is below the

primary decoder by more than the margin. These values lie in the lower tail of the permutation null (lower-tail p = 0.024 for accuracy and 0.048 for confidence), the held-out and training patterns correlate slightly negatively, and within-participant cross-validation is near chance, so the features do not transfer across participants and we identified no more specific mechanism.

The CPP-like features reached 0.509 for choice (p = 0.217), 0.508 for accuracy (p = 0.299), and 0.533 for confidence (p = 0.016); the LRP-like features reached 0.495, 0.489, and 0.514. For confidence the two CPP-like features alone came close to the whole-scalp decoder (difference +0.008, [−0.025, 0.038] in favor of the whole-scalp decoder), which is consistent with a centro-parietal contribution to the confidence effect, but this effect is itself RT-dependent (Section 5.3). For choice neither component set exceeded chance, and the whole-scalp decoder was better than the LRP-like features (+0.032, [0.006, 0.059]), so the choice information is not captured by the classical lateralized motor component alone. Adding the flexible RT basis to the component features gave AUCs equal to the RT-only model for accuracy and confidence. The window grid gave mean AUCs between 0.46 and 0.57 without a consistent optimum, so extreme cells should be read as chance variation (Supplementary Table S2).

5.6 Time-resolved and stimulus-locked decoding

Choice AUCs lay between 0.48 and 0.52 without a clear time course and no window survived Holm correction. Accuracy decoding was significant after correction in 3 windows, [-650, -550) ms, [-600, -500) ms, [-450, -350) ms (AUC 0.517 to 0.529; Supplementary Table S12), and none of these remained after flexible RT residualization. Confidence peaked at 0.541 in the window [-250, -150) ms but was not significant after correction. The time course neither supports nor refutes the physiological rationale of the 150 ms buffer: the windows are short, the signals weak, and no analysis here shows where response-related motor activity begins.

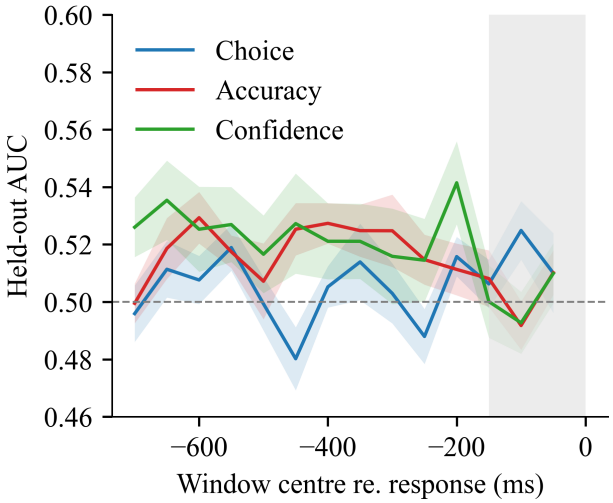


Fig. 2. Response-locked decoding in 100 ms windows. Lines show the mean held-out-participant AUC against the window centre, shaded areas the standard error across participants, and the gray band the buffer excluded from the primary analysis.

In the stimulus-locked evaluation (Supplementary Table S8), across 48 combinations of decision time (400–1100 ms), latency (0 and 150 ms), and target, mean AUCs lay between 0.47 and 0.53; only 1 combination survived Holm correction over the eight decision times (accuracy, δ = 0 ms, τ = 700 ms, AUC 0.522), and the corresponding value with a 150 ms processing latency was not significant. Choice, accuracy, and confidence therefore could not be predicted in advance from stimulus-locked windows in this dataset. This does not show that online prediction is impossible (the windows are short, the retained trial numbers fall with τ, and the dataset was not designed for it), but the response-locked AUCs of about 0.53 to 0.54 should not be read as evidence of deployable advance prediction.

5.7 What does the choice signal contain?

Stimulus direction, choice, and response preparation are coupled in this task, so the choice AUC does not identify a choice-specific representation. First, motion direction itself was decodable only weakly (AUC 0.515, permutation p = 0.061); the choice scores discriminated choice within rightward stimuli (0.516, p = 0.172) and leftward stimuli (0.520, p = 0.067) only weakly, with neither reaching significance. Second, the choice AUC was 0.534 on correct trials (choice equals stimulus; $p < 0.001$) and 0.504 on the 1,545 incorrect trials (choice opposite to the stimulus; [0.466, 0.539]; $p = 0.410$). A signal that carried only stimulus information would fall below 0.5 on incorrect trials, and one that carried only the choice would be equal in both; the incorrect-trial estimate is

too imprecise to decide between these accounts. Third, matching trials on RT and stimulus direction removed the effect (0.506, $p = 0.292$; Section 5.3). The activation pattern of the choice decoder (Haufe transform, averaged over folds and stable across them, mean pairwise correlation 0.94 [26]; Supplementary Material) correlated with the pattern of the direction decoder (r = 0.52) but only weakly with a left–right hemispheric template (r = −0.06, [−0.087, −0.040] over folds), so it is not the classical lateralized motor topography. Together these results are compatible with a weak mixture of stimulus-related, evidence-related, and choice-related information and do not exclude motor preparation. We therefore describe the surviving choice effect as weak discriminative information in this task.

5.8 RQ3: calibration

Incorrect/high trials could not be distinguished from incorrect/low trials (pooled AUC 0.488, [0.450, 0.528]; permutation $p = 0.524$; 17 evaluable participants), nor reliably from correct/high trials (0.531, [0.496, 0.576]; $p = 0.131$; 18 participants) (Table 5). Among correct trials, high versus low confidence was weakly separable (0.525, $p = 0.015$). The four-way classifier reached a balanced accuracy of 0.265 (chance 0.250; permutation $p = 0.116$) and a macro-averaged AUC of 0.517 ($p = 0.056$). In the two-stage analysis, incorrect/high trials had a slightly more correct-like accuracy-decoder score than incorrect/low trials (AUC 0.525, [0.481, 0.565]; one-sided permutation $p = 0.147$), the direction expected for an attenuated error signature but not significant, and with the score of the RT-residualized decoder the AUC was 0.497 ($p = 0.675$). Within participants, reported confidence correlated positively with the score on correct trials (mean Fisher $z = 0.079$, $p = 0.005$) and weakly on incorrect trials (0.052, $p = 0.062$); after partialling out RT ranks these correlations were 0.026 ($p = 0.274$) and 0.005 ($p = 0.857$), i.e. not distinguishable from zero. The result depends little on how the confidence classes are defined (Supplementary Table S6): among incorrect trials the high-versus-low AUC was 0.488 for ratings 1 to 3 versus 7 to 9, 0.533 for 1 to 2 versus 8 to 9 ($p = 0.059$; only 88 high-confidence errors from 7 evaluable participants), and 0.507 for 1 to 4 versus 6 to 9. The larger separation among correct than among incorrect trials is the opposite of what an attenuated pre-response error signature would predict if the EEG contained confidence information independent of accuracy.

Table 5. Calibration contrasts (pooled out-of-participant AUC; 1000 permutations)

Contrast	Trials	Pos./neg.	Eval.	AUC	95% interval	*p*
IH vs IL	853	237/616	17	0.488	[0.450, 0.528]	0.524
IH vs CH	1,866	237/1629	18	0.531	[0.496, 0.576]	0.131
IL vs CL	1,455	616/839	23	0.516	[0.492, 0.543]	0.226
CH vs CL	2,468	1629/839	24	0.525	[0.506, 0.549]	0.015

5.9 RQ4: sequential effects

There were 5,332 consecutive trial pairs from 24 participants (18 to 312 per participant); the choice was repeated on 51.1% and the motion direction on 49.1% of pairs. The decoded neural confidence state did not predict choice repetition (β = 0.019, robust SE 0.037, $p = 0.602$; bias-reduced SE 0.039, $p = 0.625$; odds ratio per SD 1.02 [0.94, 1.10]), whereas reported confidence did (β = 0.143, robust SE 0.048, $p = 0.003$; bias-reduced SE 0.050, $p = 0.009$; odds ratio 1.15 [1.04, 1.28]; Table 6). The participant bootstrap that re-fits the decoder in every resample gave an interval of [−0.054, 0.082] for the neural state (two-sided $p = 0.721$) and [0.050, 0.240] for reported confidence ($p = 0.008$); 21 and 18 of 500 resamples were discarded because the GEE did not converge. Neither predictor was reliably related to the RT or the accuracy on trial N+1 (neural state: β = 0.001 and 0.040; $p = 0.782$ and $p = 0.233$). The state obtained from the RT-residualized decoder, which carries no decodable confidence information, also showed no association (β = 0.023, $p = 0.479$).

Table 6. Predictors of trial N+1 behavior (GEE; bias-reduced *p*, 23 df)

Outcome and predictor	β	SE	*p*
Repeat: neural state *S*	0.019	0.039	0.625
Repeat: reported conf. *C*	0.143	0.050	0.009
ln RT N+1: *S*	0.001	0.002	0.782
ln RT N+1: *C*	0.004	0.004	0.235
Accuracy N+1: *S*	0.040	0.032	0.233
Accuracy N+1: *C*	0.018	0.030	0.553

5.10 Sensitivity and minimum detectable effects

The design could detect mean AUCs of about 0.525 for choice, 0.527 for accuracy, and 0.539 for confidence with 80% power (the 95th percentiles of the permutation nulls, 0.518, 0.519, and 0.528, are rejection thresholds, not power). For the calibration contrasts the pooled AUC at which the design had

80% power was 0.557 for incorrect/high versus incorrect/low, 0.562 for incorrect/high versus correct/high, 0.547 for incorrect/low versus correct/low, and 0.528 for correct/high versus correct/low. The upper bound of the 95% participant-bootstrap interval was below the smallest effect of interest (AUC 0.55) for incorrect/high versus incorrect/low (0.528) and incorrect/low versus correct/low (0.543), so effects of that size are unlikely for these contrasts, whereas the incorrect/high versus correct/high interval ([0.496, 0.576]) does not exclude it. In the sequential models the bias-reduced standard error of the neural-state coefficient for choice repetition was 0.039, so effects larger than about 1.11 in odds ratio per standard deviation would have been detected with 80% power; the reported-confidence coefficient (0.143) is above the corresponding value for that predictor (1.15). The matched choice analysis has fewer trials and a wider null distribution; its non-significance should be read as absence of detected effect, not evidence of absence, and the sequential null concerns the weak, largely RT-related decoder score only. A stronger confidence read-out or an independent replication would be needed to test whether a neural confidence influence on later behavior exists at all. Supplementary Table S11 lists all minimum detectable effects.

6. Discussion

6.1 What the pre-response signal contains

Pre-response EEG carried small but reliable information about choice, accuracy, and confidence (mean AUC 0.527 to 0.540). The accuracy and confidence effects are largely response-time information: flexible RT control removed them, RT alone predicted them better than the EEG, and adding EEG did not improve prediction. The choice effect survived RT control but not matching on RT and stimulus direction, and its scalp pattern was not the classical lateralized motor topography, so it should be called weak discriminative information in this task and not a choice-specific representation. The time-resolved analysis did not validate the physiological rationale of the buffer.

6.2 Response-time adjustment is a modelling choice

Removing an RT association is not a causal test, because RT can share the decision-generating process with the neural features and residualization can remove meaningful information. The evidence depends on the adjustment: linear and spline residualization and incremental tests agree, terciles leave a borderline accuracy effect ($p = 0.106$ by permutation) because they keep within-bin RT variation, and matching reduces the trial number and power. A stronger conclusion would need a mechanistic model or an independent confidence readout.

6.3 Retrospective decoding is not online prediction

The window ends 152 to 155 ms before the response, so its endpoint cannot be located prospectively, and the normalization needs a batch of the new participant's trials. In a stimulus-locked, causally available evaluation the effects did not reappear, with one of 48 combinations surviving correction. The AUCs of about 0.53 to 0.54 are therefore not evidence of deployable advance prediction, which would also need to account for processing latency and calibration data.

6.4 Calibration, sequential effects, and scope

We found no evidence that high-confidence errors are neurally distinguishable from low-confidence errors, and the intervals exclude an AUC of 0.55 for two of the contrasts but not for incorrect/high against correct/high. The decoded confidence state did not predict next-trial behavior, whereas reported confidence predicted choice repetition. This null concerns a weak, largely RT-related score and does not test whether a neural confidence influence on later behavior exists. The results concern slower trials of scanner EEG (17.4% of trials were excluded, retention was very uneven, and the confidence proxy is coarse), and should not be extended to fast decisions, ordinary EEG outside the scanner, or individual overconfidence detection.

7. Limitations

All relationships are predictive and correlational. The temporal isolation from response-related activity is not established for the dataset's own band-pass, and the causal re-extraction covers 9 of 24 participants. The window rule selected slower trials, the confidence proxy is a coarse split of a 9-point rating, the grid and the time-resolved analysis contain many comparisons whose extreme cells are optimistic, and the data come from one task with 24 participants. The analysis plan was internal and not preregistered.

8. Conclusion and Future Work

In 24 participants, EEG from a window of 600 ms that ended about 150 ms before the response (about −755 to −152 ms) carried weak information about choice, accuracy, and confidence (mean AUC 0.527, 0.535, 0.540). Accuracy and confidence

information did not survive flexible response-time control, and EEG did not add to response time. The choice effect survived response-time control but not joint matching on response time and stimulus direction. High-confidence errors were not separable from low-confidence errors, and the decoded confidence state did not predict next-trial behavior. The study supports weak, retrospective information in slower trials and not online prediction. Future work should split the dataset's band-pass into inside-window and later parts, use raw recordings, add a stronger confidence readout, and replicate in an independent sample.

Conflicts of Interest

The authors declare no conflict of interest.

Author Contributions

Conceptualization, A.P. and S.S.; methodology, A.P. and P.E.; software, P.E. and M.K.; validation, M.K. and P.D.; formal analysis, P.D. and S.S.; investigation, A.P. and M.K.; resources, P.E. and S.S.; data curation, P.E. and P.D.; writing—original draft preparation, A.P., P.E., M.K., P.D. and S.S.; writing—review and editing, A.P., P.E., M.K., P.D. and S.S.; visualization, M.K. and P.D.; supervision, S.S.; project administration, A.P. All authors have read and agreed to the published version of the manuscript.

Data and Code Availability

The EEG-fMRI data are publicly available at OpenNeuro (ds002739, DOI 10.18112/openneuro.ds002739.v1.0.0). This study analyzed these de-identified data and collected no new data. The analysis code, the analysis plan and amendments with file times, the environment, the channel mask, retained-trial and adjacent-pair indices, normalization and RT-regression settings, decoder configurations and seeds, participant-level predictions, machine-readable permutation statistics, and the scripts that generate every table and figure from one result file are provided as a versioned reproducibility package (release R1, with a SHA-256 manifest). GitHub repo link: https://github.com/pranave-2/BCI-Project/tree/main. The editor encourages a public release. The institute doesn't require an ethics statement for secondary analysis of public, de-identified data.

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